CPU

Datei: CPU.kicad_sch

Video−Generation

Datei: Video−Generation.kicad_sch

Color−Modulator

Datei: Color−Modulator.kicad_sch

RF−Stage VHF

Datei: RF−Stage VHF.kicad_sch

RF−Stage UHF

Datei: RF−Stage UHF.kicad_sch

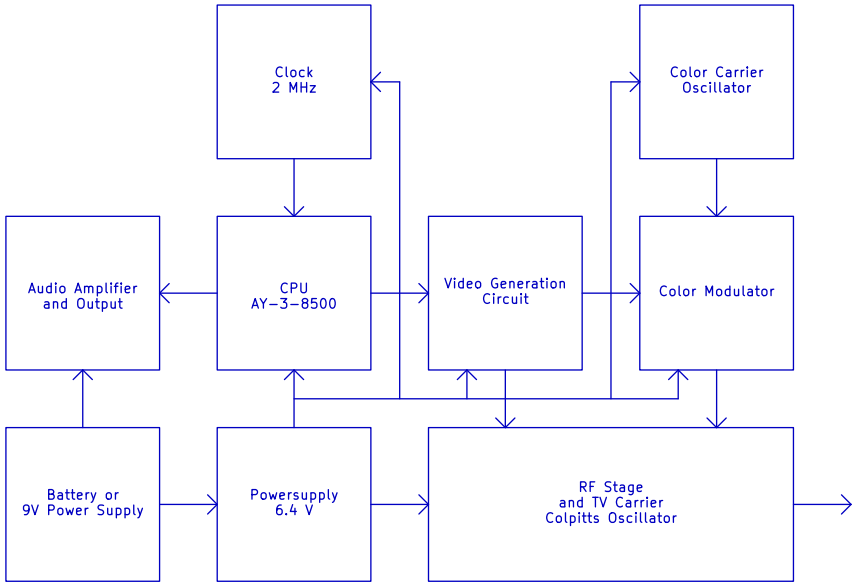
Symbol	Note
7Vrms 1	Measurement point with measured value
1	Adjustment point
1	Curve shape (oscillogram)
1	
1	Component or Circuit Note
	Module/Cassette
	PCB
	Functional group

Note regarding component names:

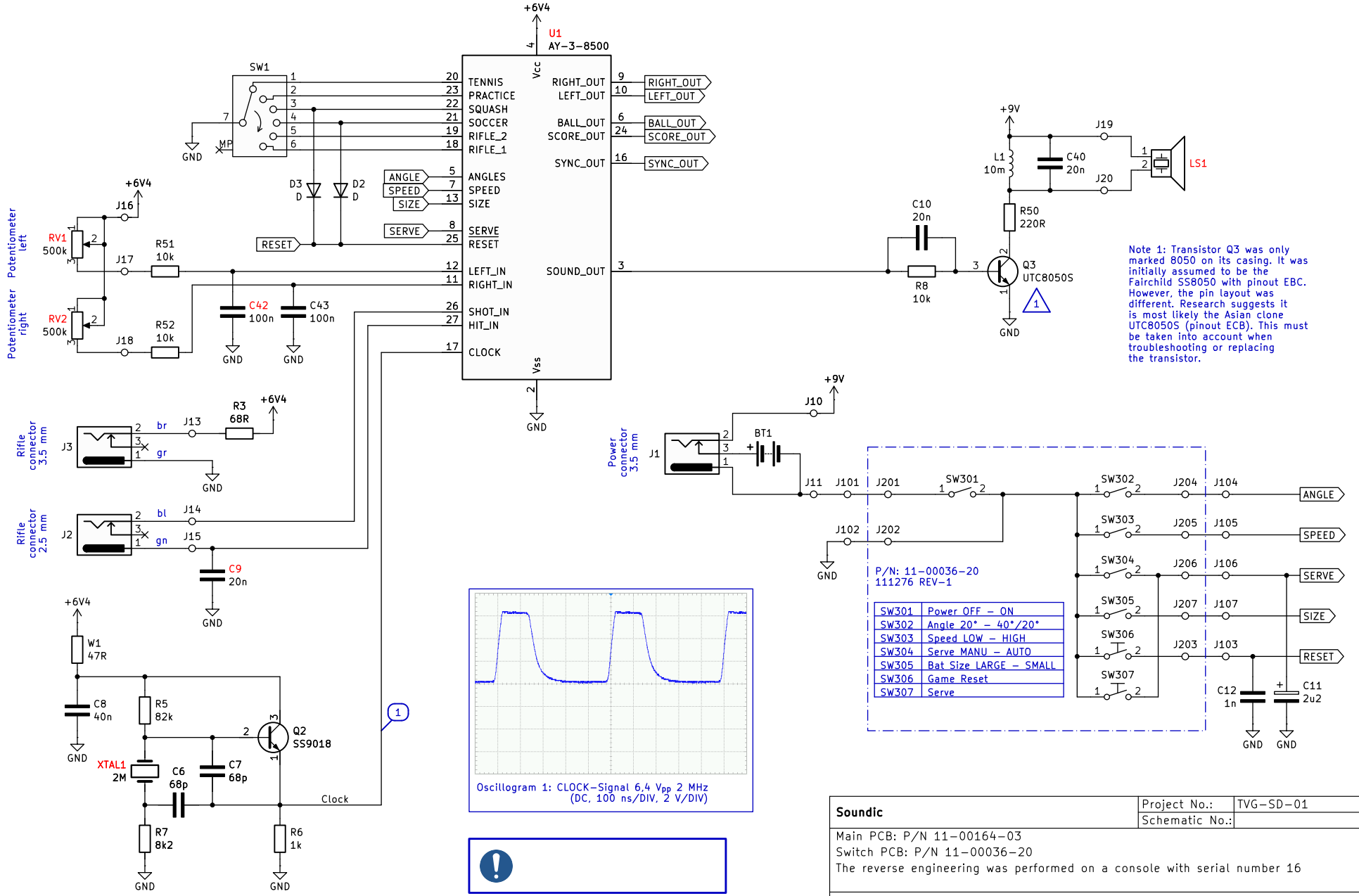
All component designations written in red were either missing, illegible, or duplicated. I attempted to deduce their meaning logically and have noted below which designations this applies to.

- C1 assigned because it is unreadable
- C4 assigned because it is unreadable
- C9 assigned because there is no label
- C42 assigned because it is unreadable
- C101 assigned because it is unreadable
- C106 assigned because it is unreadable
- D21 assigned because it is duplicated (D20)
- FL1, FL2, FL3 assigned because they are unreadable
- Jx assigned because there is no label
- L101 assigned because it is unreadable
- LS1 assigned because there is no label
- Q101 assigned because it is unreadable
- RV1, RV2 assigned because there are no labels
- U1, U2, U3 assigned because they are unreadable
- XTAL1 assigned because it is unreadable

All measurements with the oscilloscope were taken with a 1:10 probe to avoid loading the RF circuits.



Soundic				Project No.:	TVG−SD−01
				Schematic No.:	
Main PCB: P/N 11−00164−03 Switch PCB: P/N 11−00036−20 The reverse engineering was performed on a console with serial number 16					
Title: Pong Game Console Color (PAL)					
Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026−10−30
Sheet:	/			Rev:	1.0
File:	kicad_schematic.kicad_sch			Page:	1/6

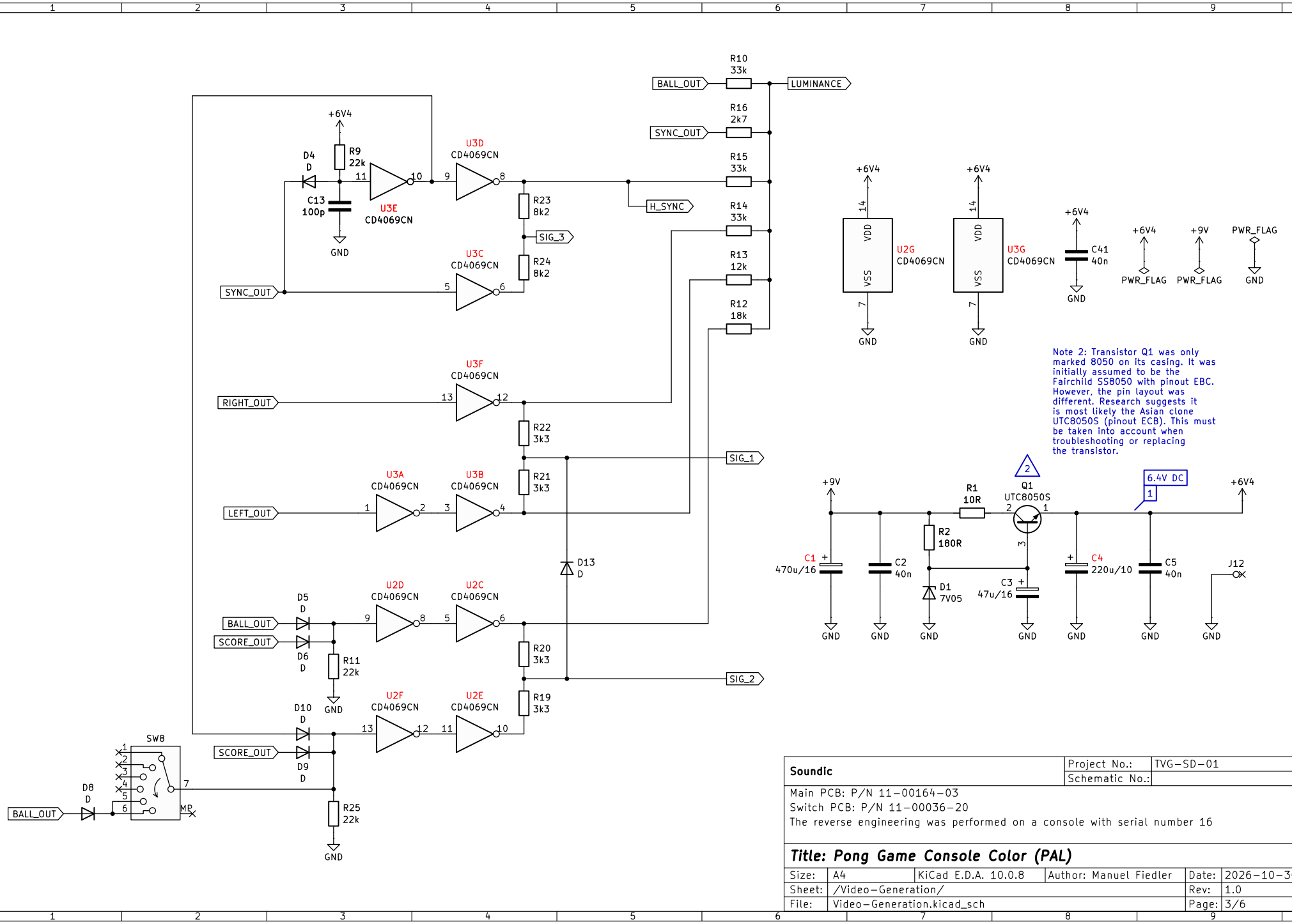


Soundic		Project No.:	TVG-SD-01
		Schematic No.:	

Main PCB: P/N 11-00164-03
Switch PCB: P/N 11-00036-20
The reverse engineering was performed on a console with serial number 16

Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026-10-30
Sheet:	/CPU/			Rev:	1.0
File:	CPU.kicad_sch			Page:	2/6



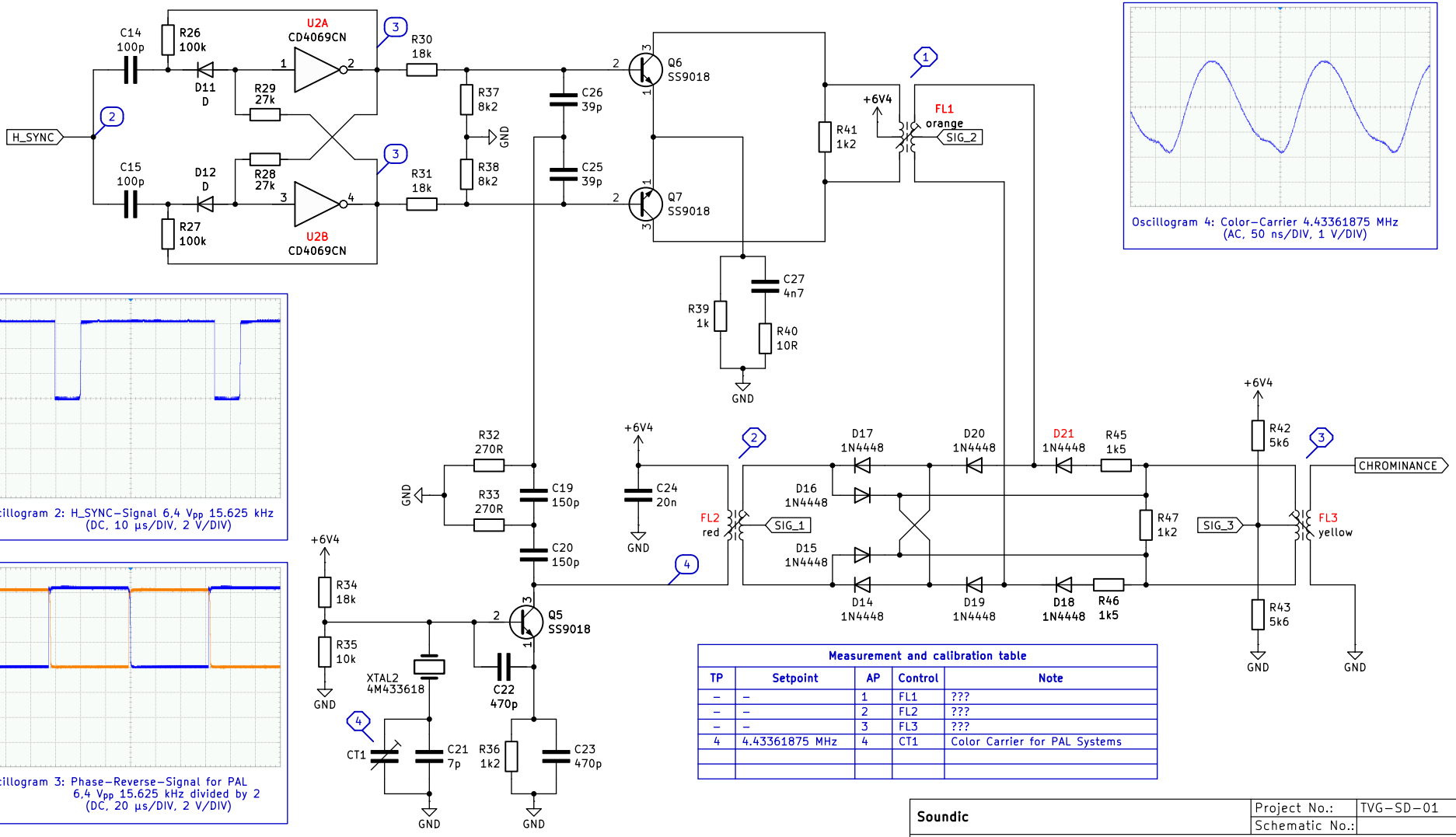
Soundic	Project No.:	TVG-SD-01	
	Schematic No.:		

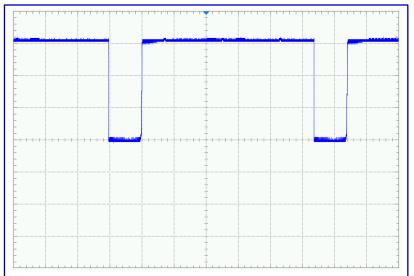
Main PCB: P/N 11-00164-03
Switch PCB: P/N 11-00036-20
The reverse engineering was performed on a console with serial number 16

Title: Pong Game Console Color (PAL)

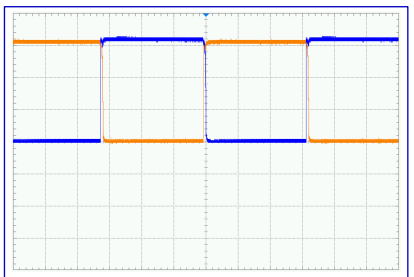
Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026-10-30
Sheet:	/Video-Generation/			Rev:	1.0
File:	Video-Generation.kicad_sch			Page:	3/6

T-Flip-Flop
Frequency Divider





Oscillogram 2: H_SYNC-Signal 6,4 V_{pp} 15.625 kHz
(DC, 10 µs/DIV, 2 V/DIV)



Oscillogram 3: Phase-Reverse-Signal for PAL
6,4 V_{pp} 15.625 kHz divided by 2
(DC, 20 µs/DIV, 2 V/DIV)

Oscillogram 4: Color-Carrier 4.43361875 MHz
(AC, 50 ns/DIV, 1 V/DIV)

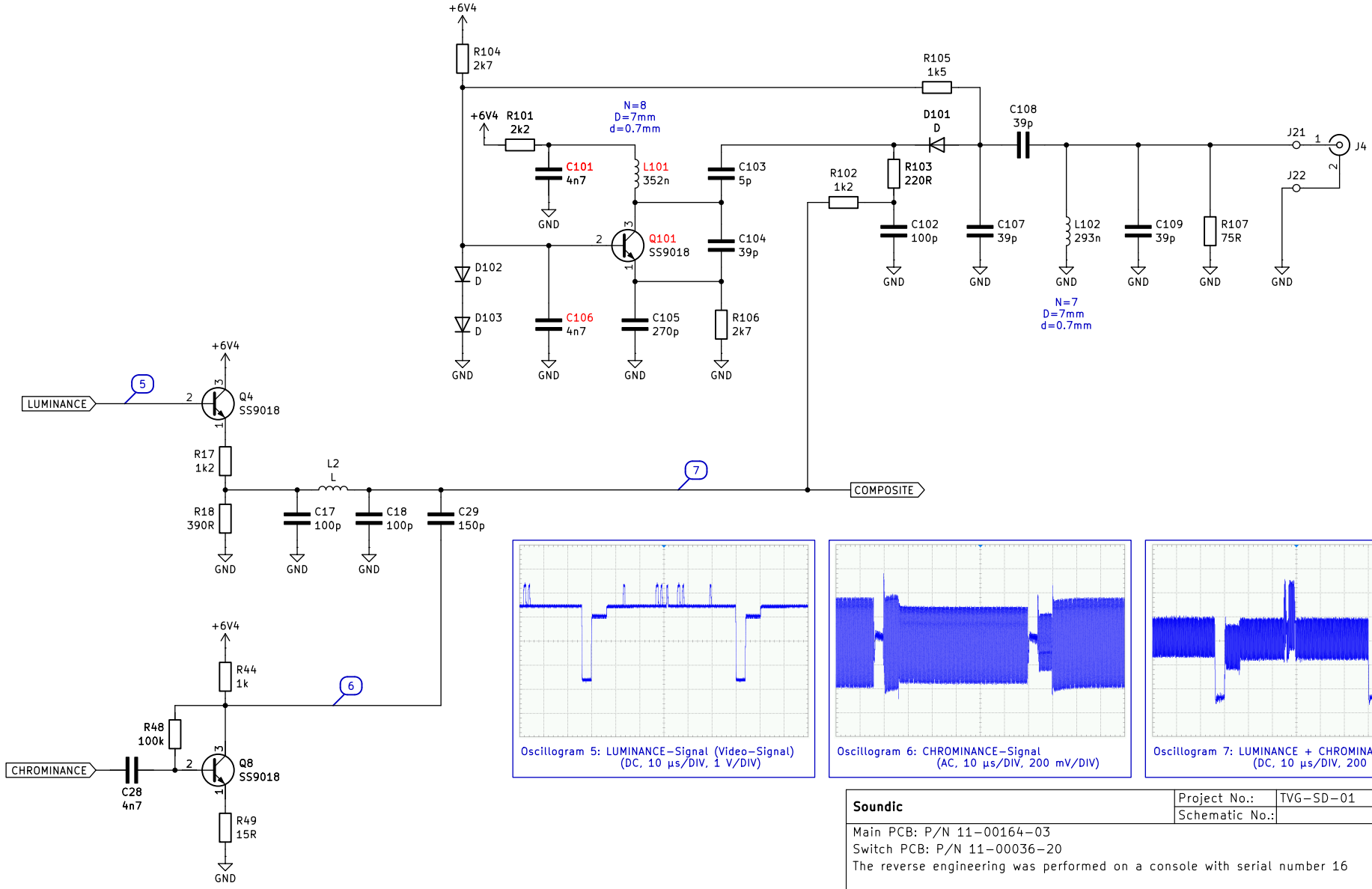
Measurement and calibration table				
TP	Setpoint	AP	Control	Note
–	–	1	FL1	???
–	–	2	FL2	???
–	–	3	FL3	???
4	4.43361875 MHz	4	CT1	Color Carrier for PAL Systems

Soundic	Project No.:	TVG-SD-01	
	Schematic No.:		

Main PCB: P/N 11-00164-03
Switch PCB: P/N 11-00036-20
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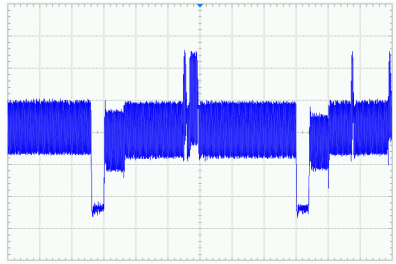
Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026-10-30
Sheet:	/Color-Modulator/			Rev:	1.0
File:	Color-Modulator.kicad_sch			Page:	4/6



Oscillogram 5: LUMINANCE–Signal (Video–Signal)
(DC, 10 µs/DIV, 1 V/DIV)

Oscillogram 6: CHROMINANCE–Signal
(AC, 10 µs/DIV, 200 mV/DIV)



Oscillogram 7: LUMINANCE + CHROMINANCE–Signal
(DC, 10 µs/DIV, 200 mV/DIV)

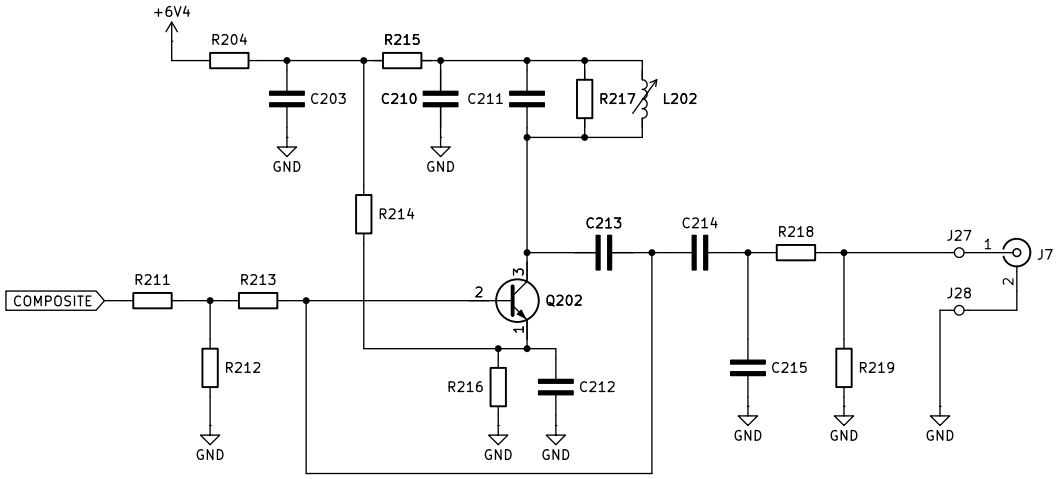
Soundic		Project No.:	TVG–SD–01
		Schematic No.:	

Main PCB: P/N 11–00164–03
Switch PCB: P/N 11–00036–20
The reverse engineering was performed on a console with serial number 16

Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026–10–30
Sheet:	/RF–Stage VHF/			Rev:	1.0
File:	RF–Stage VHF.kicad_sch			Page:	5/6

The UHF stage was created based on the component labels on the circuit board. This section is unpopulated in my console, so I cannot provide component values or guarantee the accuracy of the circuit.



Soundic		Project No.:	TVG-SD-01
		Schematic No.:	

Main PCB: P/N 11-00164-03
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Title: Pong Game Console Color (PAL)

Size:	A4	KiCad E.D.A. 10.0.8	Author: Manuel Fiedler	Date:	2026-10-30
Sheet:	/RF-Stage UHF/			Rev:	1.0
File:	RF-Stage UHF.kicad_sch			Page:	6/6